

AI Supply Chain Control Tower - Automated Risk Analysis & Nearshoring Agent built with Antigravity AI

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1. Summary

- This project demonstrates the implementation of a No-Code AI Agent using Antigravity AI to automate the evaluation and risk management of a global supply chain. Leveraging my background in operational logistics, transportation management, I designed a workflow that autonomously identifies supply bottlenecks and suggests strategic interventions, acting as a human-in-the-loop decision support system.

2. Tech Stack & Methodology

- Data Management: SQL-based logic applied to structured KPI datasets (CSV)
- AI Engine: Antigravity AI Agent (Review-driven development mode), Prompt engineering
- Business Frameworks: Lean Management (Waste reduction in decision-making)
- Role: AI Orchestrator & Logistics Subject Matter Expert

3. The Problem Statement

- Manual monitoring of supplier KPIs (like On-Time In-Full deliveries and Price Volatility) across multiple regions is time-consuming. In a dynamic logistics landscape, a proactive system is needed to flag underperforming suppliers before they cause stockouts at the retail level.

4. The AI Workflow & Architecture

- I configured the AI Agent to function as a Strategic Procurement Assistant. The operational logic includes:
- Data Ingestion: Connecting the agent to a dataset of 40 global suppliers.
- Rule-Based Flagging: Establishing strict thresholds (e.g., OTIF < 85% triggers a Critical Risk alert).
- AI Reasoning: The agent analyzes the intersection of poor delivery times, high costs, and regional transit risks (e.g., Asia vs. Europe lead times).
- Human-in-the-Loop: Operating in 'Review-driven' mode ensures that all AI-generated mitigation strategies (such as dual-sourcing or safety stock adjustments) are validated by a human logistics expert before approval.

5. Key Outcomes & Future Scope

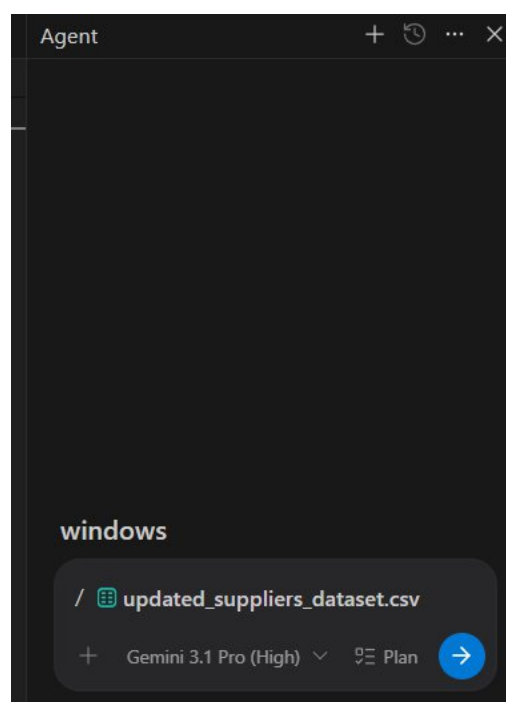
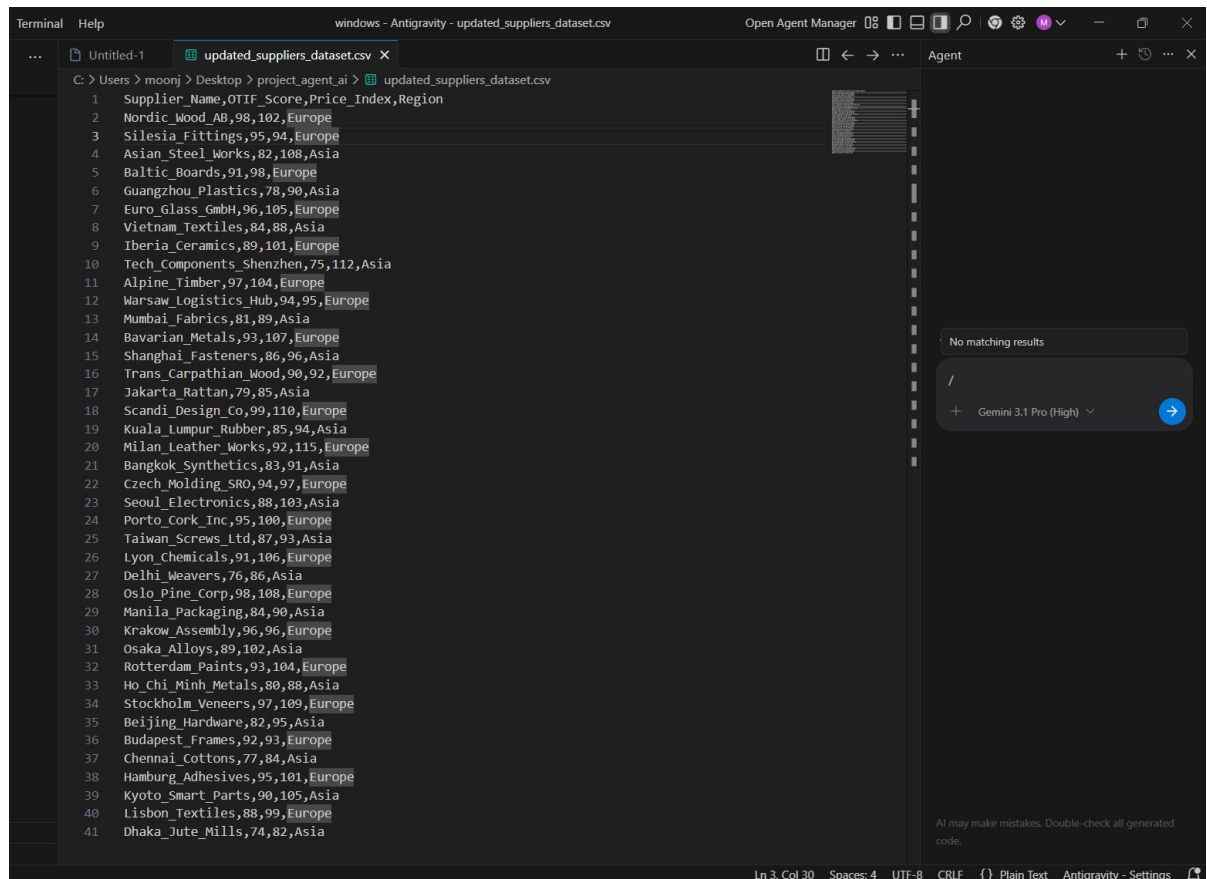
- Successfully automated the initial supplier screening process, drastically reducing the time needed for risk analysis.
- Proved the viability of integrating AI Agents as "Bolt-on" solutions to traditional ERP systems.
- Exploring integration with real-time transport visibility platforms aligning with Lean Management optimization goals.

6. Workflow execution

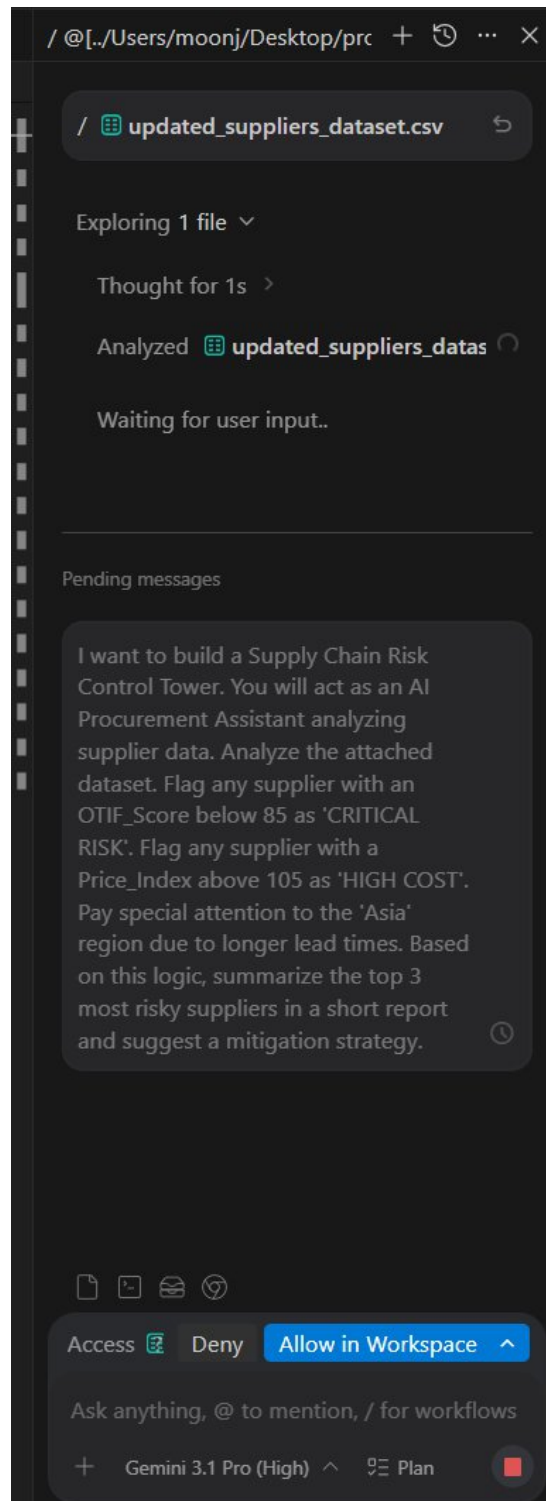
- Workflow execution represents the system's transition from analysis to action.

6.1. Data ingestion and setting business logic

To begin the process, I uploaded the *updated_suppliers_dataset.csv* and initialized the AI Procurement Assistant. I defined the core risk parameters: flagging 'CRITICAL RISK' for OTIF scores below 85, 'HIGH COST' for Price Indexes above 105, and emphasizing regional vulnerabilities in Asia.



'I want to build a Supply Chain Risk Control Tower. You will act as an AI Procurement Assistant analyzing supplier data. Analyze the attached dataset. Flag any supplier with an OTIF_Score below 85 as 'CRITICAL RISK'. Flag any supplier with a Price_Index above 105 as 'HIGH COST'. Pay special attention to the 'Asia' region due to longer lead times. Based on this logic, summarize the top 3 most risky suppliers in a short report and suggest a mitigation strategy.'



6.2. Automated risk report generation

The Agent successfully analyzed the data and generated a comprehensive report. It identified critical vulnerabilities and proposed initial mitigation strategies.

Worked for 40s >

I've analyzed the supplier dataset according to your parameters and created a detailed report.

I identified two suppliers that hit all three risk flags (Critical Risk OTIF, High Cost, and located in Asia), plus one that has an extremely concerning OTIF score.

I've documented the findings and my suggested mitigation strategy in the `supplier_risk_report.md` artifact that you can review. Let me know if you would like me to adjust any of the thresholds or perform any further analysis on this data!

 Supplier Risk Report

5:28 PM



Supplier Risk Report 1 minute ago

 Review

Supply Chain Risk Control Tower: Supplier Analysis

Based on the provided dataset, the suppliers have been evaluated using the following criteria:

- **CRITICAL RISK:** OTIF_Score < 85
- **HIGH COST:** Price_Index > 105
- **Regional Risk:** Suppliers in Asia have an elevated risk profile due to longer lead times.

 Top 3 Most Risky Suppliers

The following suppliers represent the highest risk to the supply chain based on the compounded risk factors:

Supplier Name	OTIF Score	Price Index	Region	Risk Flags
Tech_Components_Shenzhen	75	112	Asia	 CRITICAL RISK ,  HIGH COST ,  ASIA
Asian_Steel_Works	82	108	Asia	 CRITICAL RISK ,  HIGH COST ,  ASIA
Dhaka_Jute_Mills	74	82	Asia	 CRITICAL RISK ,  ASIA

WARNING

Tech_Components_Shenzhen and **Asian_Steel_Works** both trigger all three risk parameters: they suffer from poor delivery reliability (CRITICAL RISK), are excessively expensive (HIGH COST), and are located in a high-lead-time region (Asia).

Dhaka_Jute_Mills holds the lowest overall OTIF score in the dataset (74).

Suggested Mitigation Strategy

To address these vulnerabilities, the following mitigation strategies are recommended:

1. Immediate Engagement & RCA (Root Cause Analysis)

- Initiate immediate performance reviews with the identified suppliers. Demand a corrective action plan (CAP) specifically addressing their low OTIF scores.

2. Dual-Sourcing / Nearshoring

- Given the combined "High Cost" and "Asia" regional risk, begin identifying backup suppliers in the "Europe" region for critical components currently sourced from *Tech_Components_Shenzen* and *Asian_Steel_Works*. European alternatives present in the dataset (e.g., *Silesia_Fittings*, *Warsaw_Logistics_Hub*) show strong OTIF scores and lower price indices.

3. Renegotiate or Phase Out

- Use the low OTIF scores as leverage to renegotiate the high Price Indexes (above 105). If performance does not improve within a defined quarter, initiate a phased exit strategy to transition volume to more reliable and cost-effective suppliers.

6.3. Executing Nearshoring Strategy

In accordance with agile methodology principles, our goal is to maintain continuous delivery. To mitigate the identified risks in the Asian supply chain, I directed the Agent to execute a nearshoring strategy by finding European backups with high OTIF (>90) and lower costs (<100). The Agent instantly secured two viable alternatives.

‘Let's execute step 2 (Dual-Sourcing/Nearshoring). Please scan the dataset again and recommend exactly 2 specific suppliers from the 'Europe' region that have an OTIF_Score > 90 and a Price_Index < 100. They will be our backup for Asian_Steel_Works.’

Let's execute step 2 (Dual-Sourcing/Nearshoring). Please scan the dataset again and recommend exactly 2 specific suppliers from the 'Europe' region that have an OTIF_Score > 90 and a Price_Index < 100. They will be our backup for Asian_Steel_Works.

Worked for 24s >

I've scanned the dataset and identified two excellent candidates from Europe to serve as backups for Asian_Steel_Works. I have updated the `supplier_risk_report.md` artifact to include these recommendations:

- **Krakow_Assembly**: OTIF Score of 96, Price Index of 96.
- **Silesia_Fittings**: OTIF Score of 95, Price Index of 94.

Both of these suppliers easily pass the threshold of an OTIF Score > 90 and a Price Index < 100, providing much better reliability and significant cost savings. Let me know if you need any further analysis!



Suggested Mitigation Strategy

To address these vulnerabilities, the following mitigation strategies are recommended:

1. Immediate Engagement & RCA (Root Cause Analysis)

- Initiate immediate performance reviews with the identified suppliers. Demand a corrective action plan (CAP) specifically addressing their low OTIF scores.

2. Dual-Sourcing / Nearshoring

- Given the combined "High Cost" and "Asia" regional risk, we are executing a nearshoring strategy to secure backup suppliers in the "Europe" region for critical components currently sourced from *Asian_Steel_Works*.
- **Recommended European Backups:**
 - **Krakow_Assembly:** OTIF Score of 96, Price Index of 96.
 - **Silesia_Fittings:** OTIF Score of 95, Price Index of 94.
- Both of these suppliers meet the strict criteria (OTIF > 90, Price Index < 100), offering significantly better reliability and immediate cost efficiency compared to the baseline.

3. Renegotiate or Phase Out

- Use the low OTIF scores as leverage to renegotiate the high Price Indexes (above 105). If performance does not improve within a defined quarter, initiate a phased exit strategy to transition volume to more reliable and cost-effective suppliers.

6.4. Decision Workflow Visualization

To finalize the project documentation, the AI generated a visual representation of the logic executed: filtering global data, identifying risk nodes, and applying the nearshoring strategy.

'Please generate a flowchart diagram illustrating the decision-making process we just completed. The flowchart should show the workflow: starting from 'Global Supplier Data', passing through 'Risk Filtering (OTIF < 85, Price Index > 105, Region: Asia)', highlighting the 'High Risk Identified (Asian_Steel_Works)', and ending with 'Execution: Nearshoring to Europe (Krakow_Assembly & Silesia_Fittings).'

Please generate a flowchart diagram illustrating the decision-making process we just completed. The flowchart should show the workflow: starting from 'Global Supplier Data', passing through 'Risk Filtering (OTIF < 85, Price Index > 105, Region: Asia)', highlighting the 'High Risk Identified (Asian_Steel_Works)', and ending with 'Execution: Nearshoring to Europe (Krakow_Assembly & Silesia_Fittings)'.

Worked for 26s >

I have appended the requested flowchart to the bottom of the `supplier_risk_report.md` artifact. It uses Mermaid to visualize the decision-making process, mapping the path from global data ingestion through risk filtering, identifying our target high-risk supplier, and concluding with the nearshoring execution plan!

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Decision workflow Visual representation of the logic executed by the AI Agent: filtering global data, identifying risk nodes, and applying the nearshoring strategy:

